



NATIONAL UNIVERSITY
of Computer & Emerging Sciences

Software Engineering

The Smart Retail Demand Forecasting System

RetailIQ

Submitted by:

Ayesha Khalid (23L-0667)

Uma E Rubab (23L-0928)

Submitted to:

Ms. Ansa Liaqut

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Smart Retail Demand Forecasting System

(RetailIQ)

1. Project Overview

RetailIQ is a web-based retail analytics platform designed to help businesses forecast product demand and understand the factors influencing sales performance. The system uses historical retail sales data to generate demand forecasts and provide meaningful insights through interactive dashboards and analytical tools.

Traditional retail forecasting tools only predict future sales numbers but fail to explain the reasons behind demand fluctuations. RetailIQ improves this by providing interpretable analysis of sales patterns, promotion impacts, and product relationships. The platform allows users to upload sales datasets, visualize sales trends, analyze promotional effects, and receive business recommendations based on data-driven insights.

The system integrates machine learning models with a user-friendly web interface to help retail managers make informed decisions regarding inventory management, promotions, and product stocking strategies. Through interactive visualizations and analytical dashboards, RetailIQ transforms complex retail data into actionable insights.

2 User Stories

Sprint 2 focuses on building and evaluating prediction capabilities to provide accurate sales forecasts for better business decision-making.

Story ID: US-01Story Title: Access to Sales Predictions	Priority: High Estimated Hours: 5
User Story: As a retail analyst I want the system to generate sales predictions So that I can estimate future sales trends and plan accordingly Description: This feature allows users to view predicted sales values based on historical data to support forecasting. Sub User Stories: • I want to view predicted sales values for upcoming dates • I want predictions based on historical sales data • I want predictions to update when new data is available . Acceptance Criteria: • Given historical sales data is available When the user requests predictions Then the system should display future sales values • Given new data is added When predictions are generated again Then updated predictions should be shown	

Story ID: US-02 Story Title: Model Accuracy Insights	Priority: High Estimated Hours: 7
User Story: As a retail manager I want to see prediction accuracy So that I can trust the forecasts Description: This feature provides simple and clear insights into how accurate the predictions are. Sub User Stories: • I want to view accuracy metrics in a simple format • I want to compare performance of different prediction approaches • I want visual summaries of prediction accuracy Acceptance Criteria: • Given predictions are generated When the user views performance Then the system should display accuracy metrics • Given multiple prediction methods exist When the user checks performance Then a comparison view should be shown	

Story ID: US-03 Story Title: Improved Prediction Quality	Priority: Medium Estimated Hours: 6
User Story: As a business user I want the system to provide highly accurate predictions So that I can minimize risks in planning inventory and sales strategies Description: This story involves generating analytical insights such as time-based patterns, trends, and rolling averages to help users understand business performance. Sub User Stories: • I want predictions to improve over time • I want the system to reduce forecasting errors • I want consistent and reliable prediction results Acceptance Criteria: • Given predictions are generated When the system evaluates results Then prediction errors should be minimized • Given system improvements are applied When predictions are generated again Then accuracy should improve or remain consistent	

Story ID: US-04 Story Title: Reliable Forecasting System	Priority: High Estimated Hours: 5
User Story: As a retail decision-maker I want the system to use the best available prediction method So that I receive the most reliable sales forecasts Description: This ensures the system consistently uses the most effective method to provide reliable results. Sub User Stories: • I want the system to automatically use the most accurate model • I want consistent prediction results across different scenarios • I want confidence that the system is using optimized methods Acceptance Criteria: • Given multiple prediction methods are available When predictions are generated Then the system should use the most accurate method • Given predictions are shown When the user views results Then outputs should be consistent and reliable	

3 Scrum Board

Backlog for Sprint 2:

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Timeline

Backlog

Board

Calendar

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Linear Regression Model Testing

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Training of LightGBM and XGBOOST

RET-11

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Testing of LightGBM and XGBOOST

RET-12

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Fine Tuning

RET-13

8

IN PROGRESS 2

Sprint 2 - Model Traning

RET-6

Linear Regression Model Training

RET-9

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DONE

Summary

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IN PROGRESS 3

Sprint 2 - Model Traning

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Training of LightGBM and XGBOOST

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Testing of LightGBM and XGBOOST

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Linear Regression Model Training

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Linear Regression Model Testing

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RetailIQ

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Summary

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Sprint 2 - Model Traning

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Linear Regression Model Training

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Linear Regression Model Testing

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Training of LightGBM and XGBOOST

RET-11

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Testing of LightGBM and XGBOOST

RET-12

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Fine Tuning

RET-13

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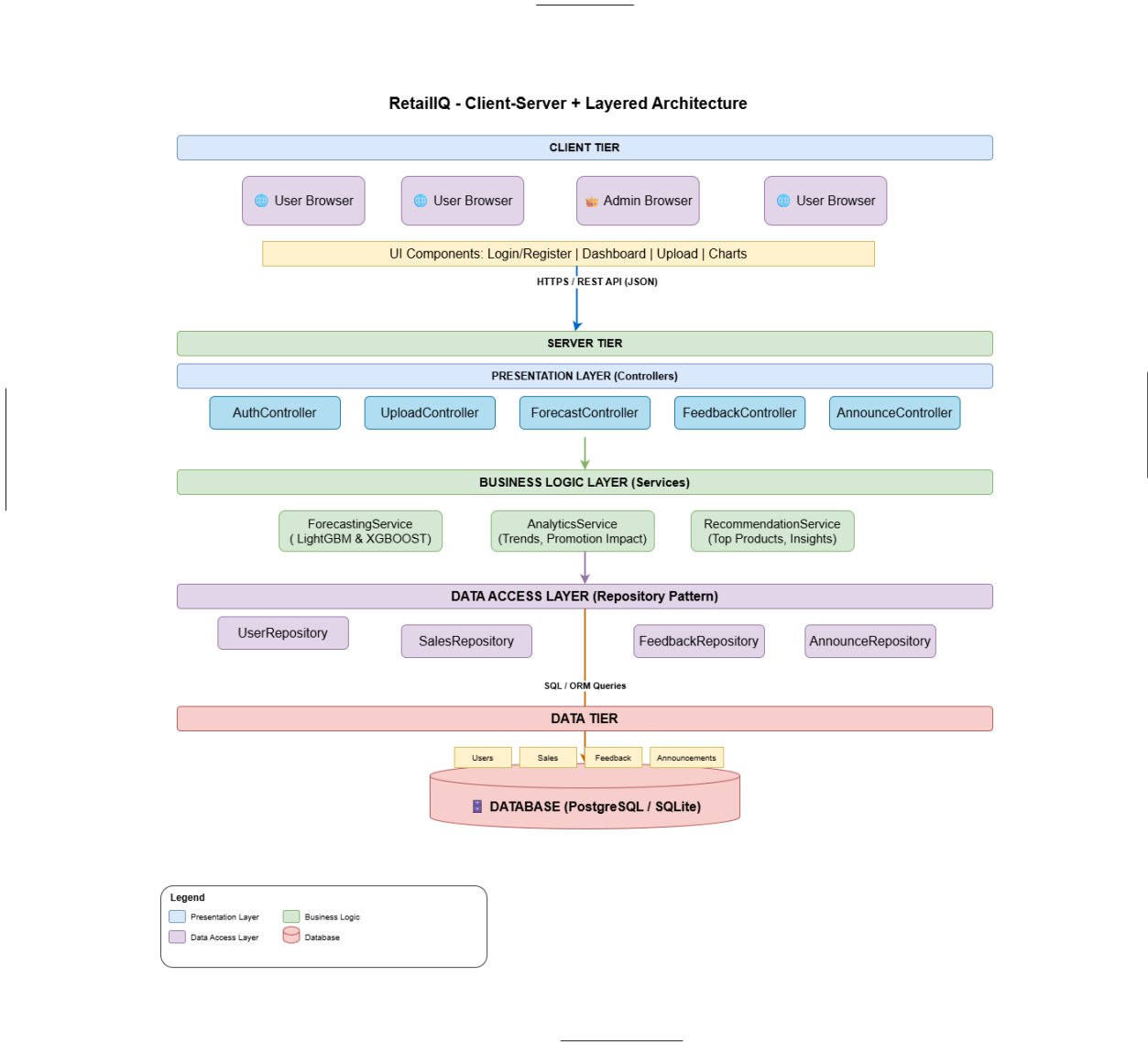
8

Work Finished

7	+	8	9	10
RetailIQ Sprint 2				
SPRINT ... × CLOSED RetailIQ Sprint 2 Start date Apr 07, 2026 Complete date Apr 11, 2026				
		15	16	17
		22	23	24

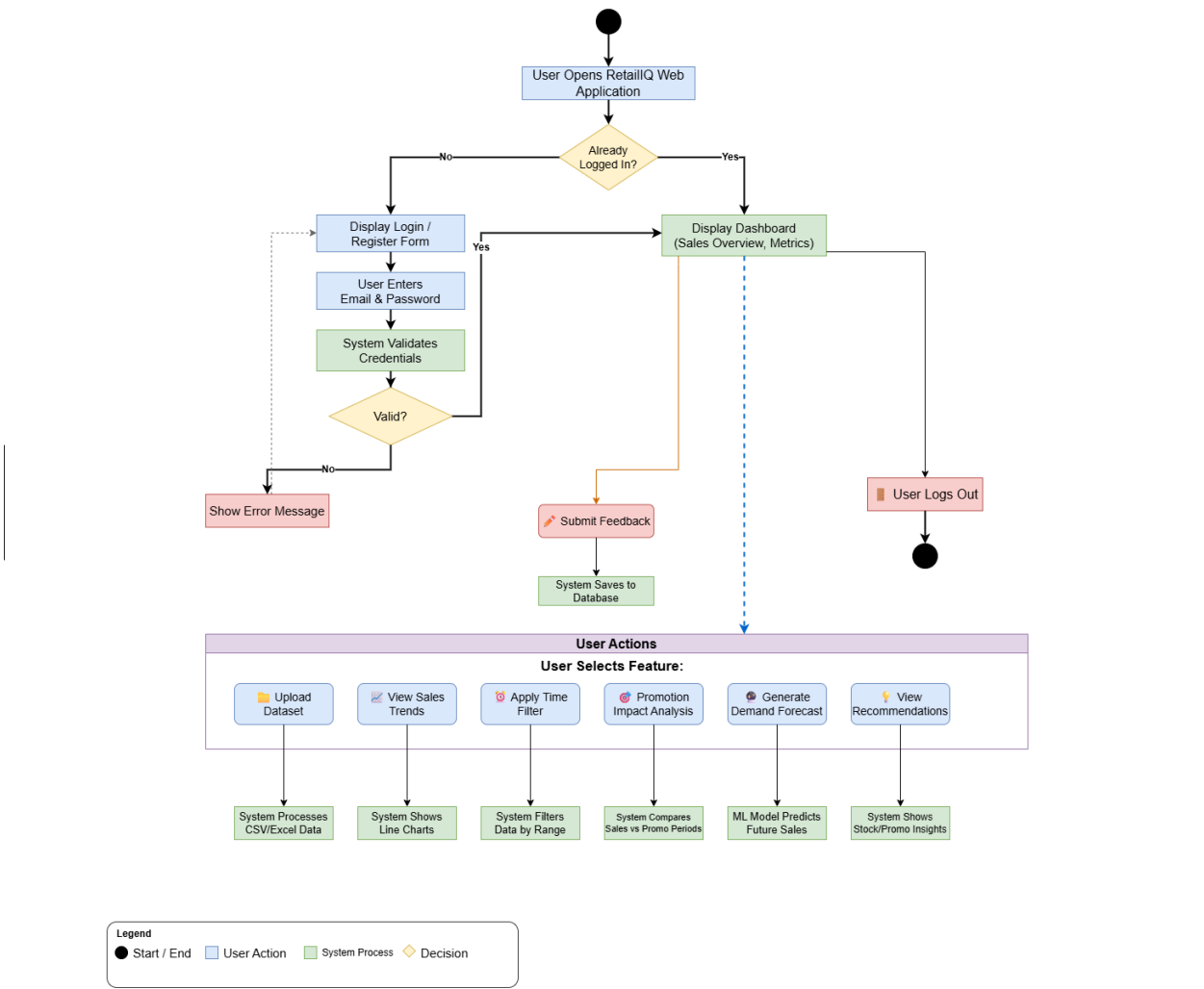
4 Diagrams

Architecture Diagram:

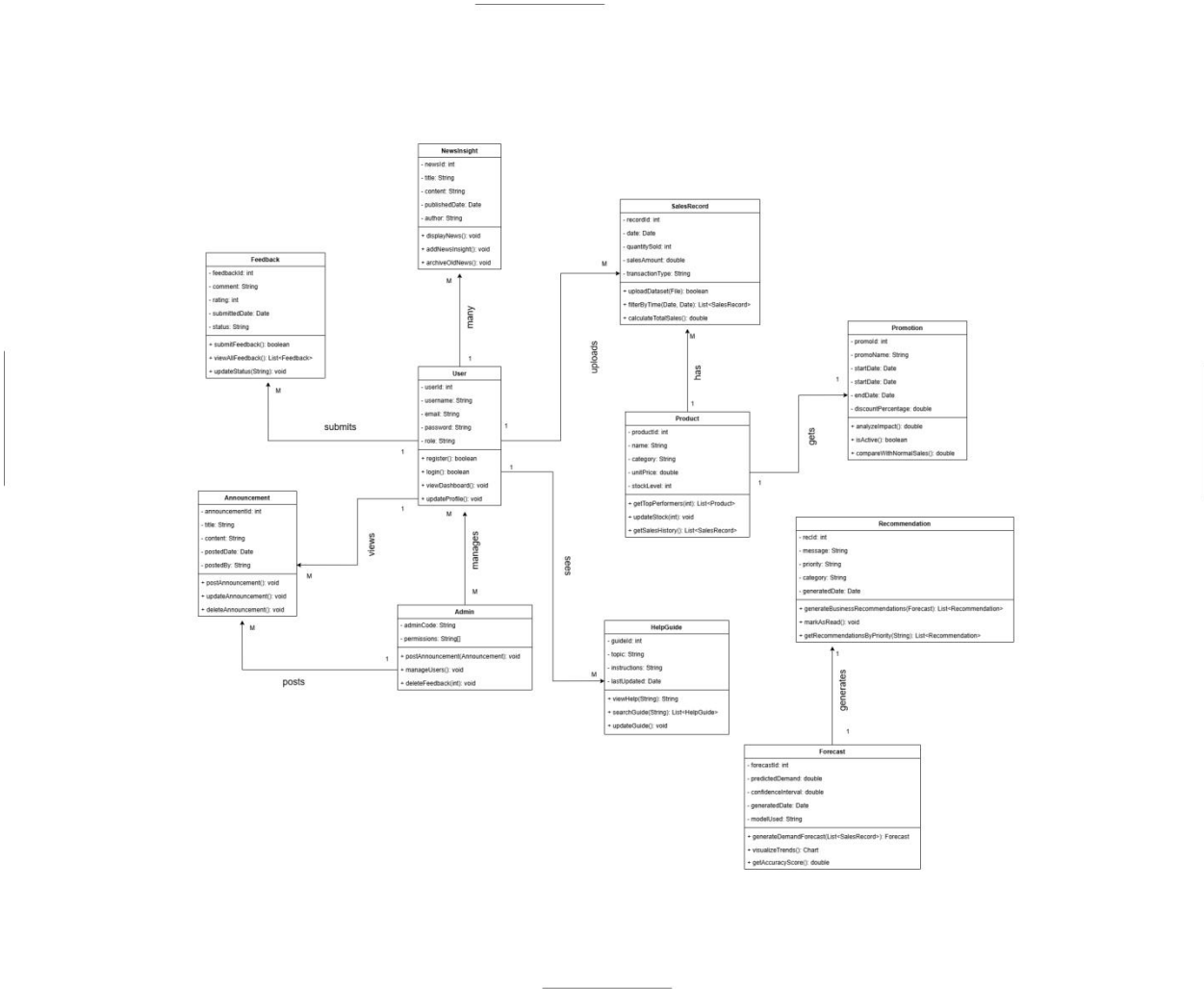


Activity Diagram

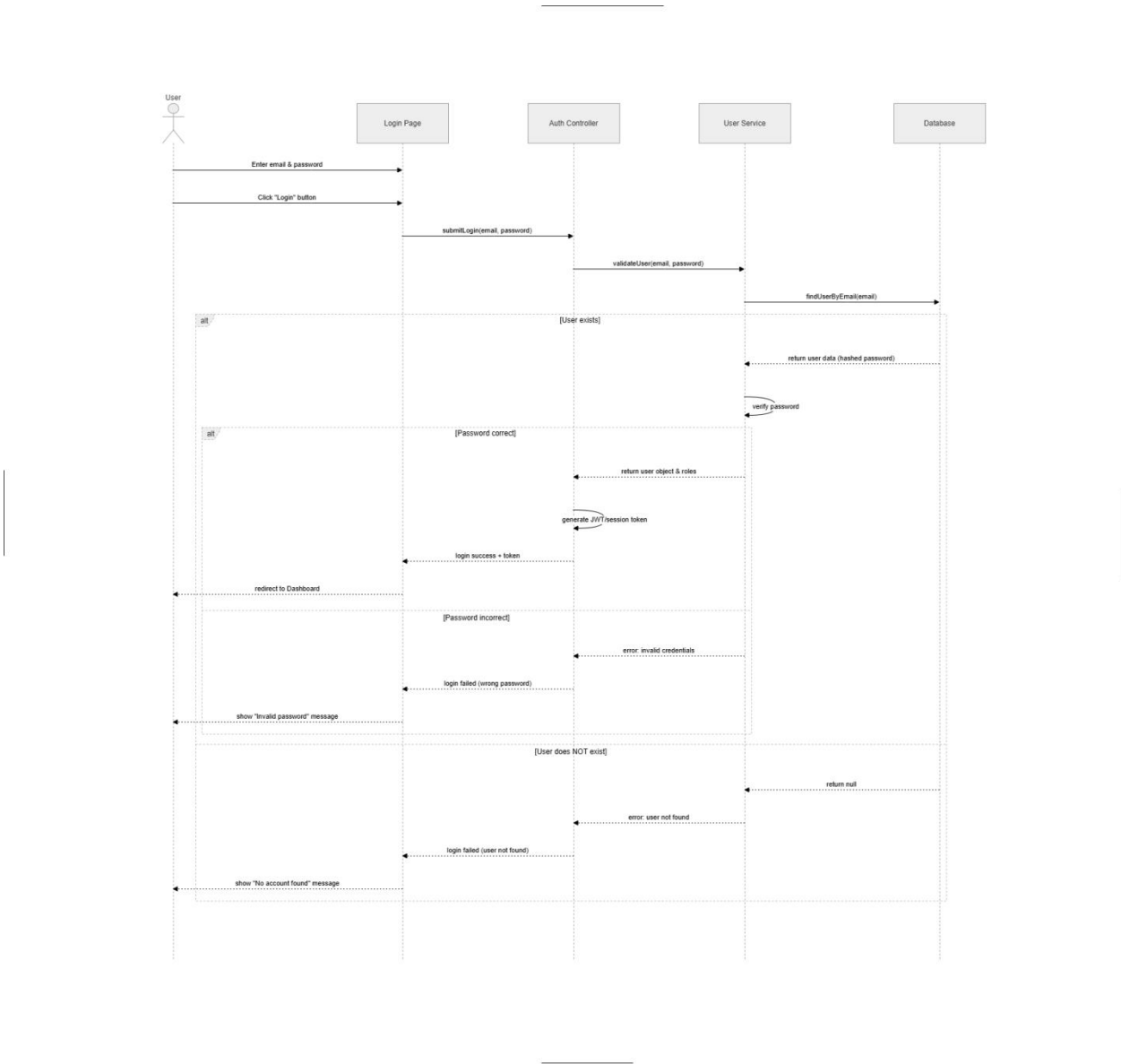
RetailIQ - Activity Diagram (User Workflow)



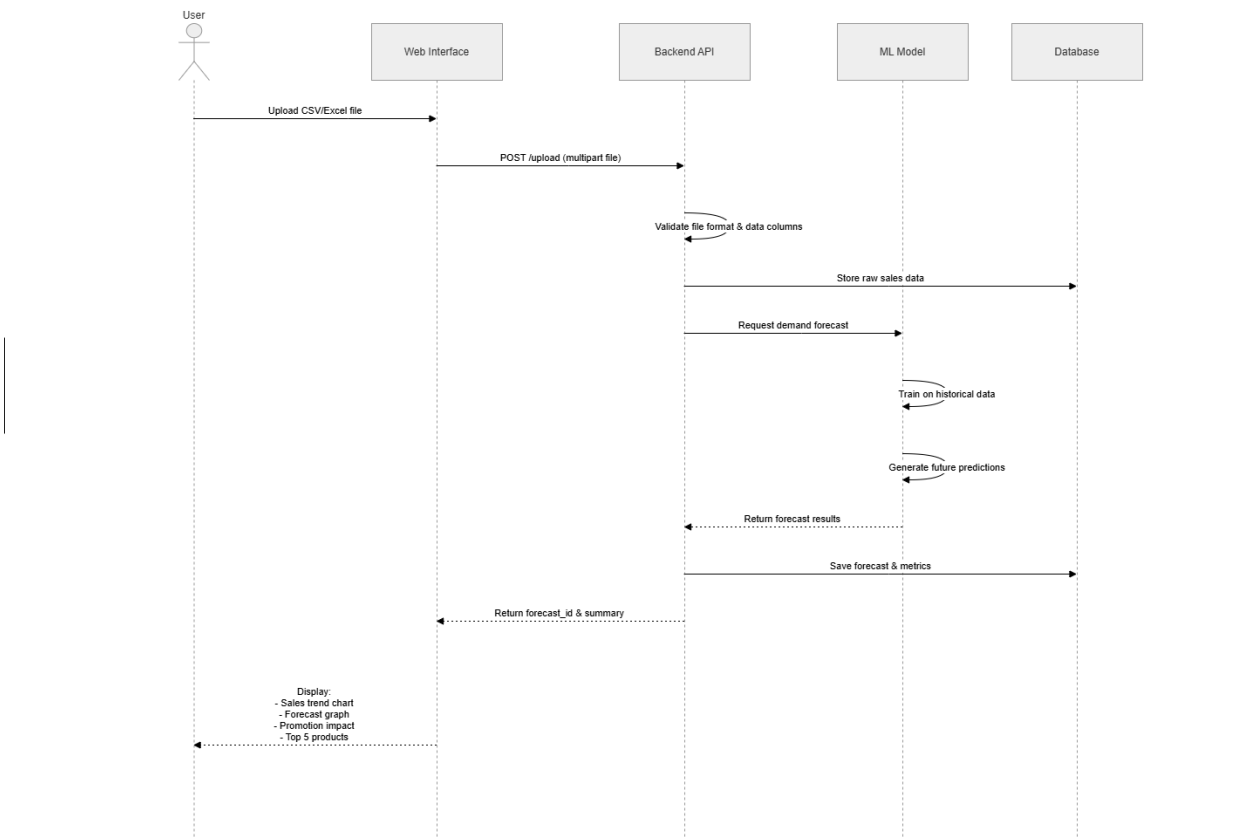
Class Diagram



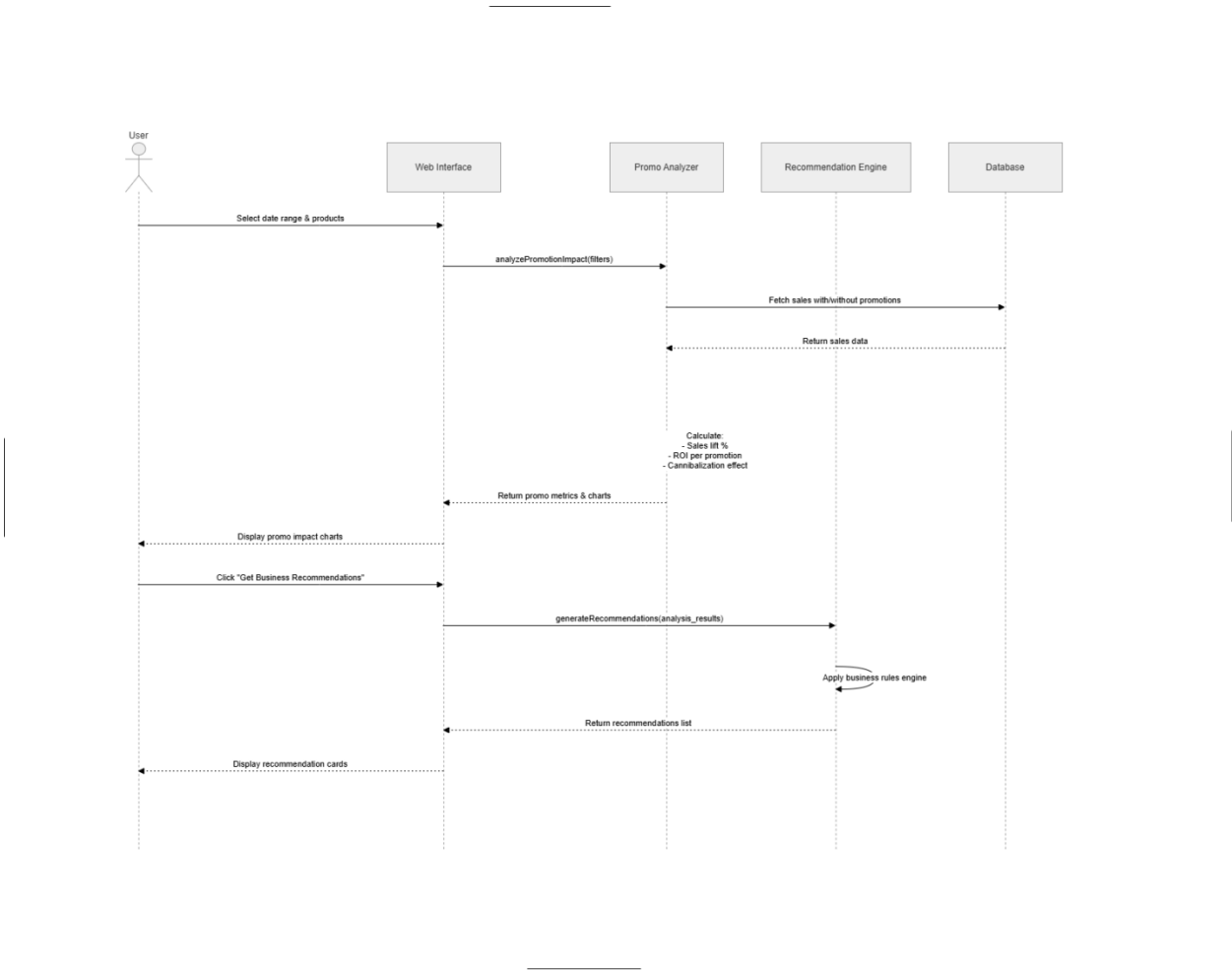
Sequence Diagram from Login



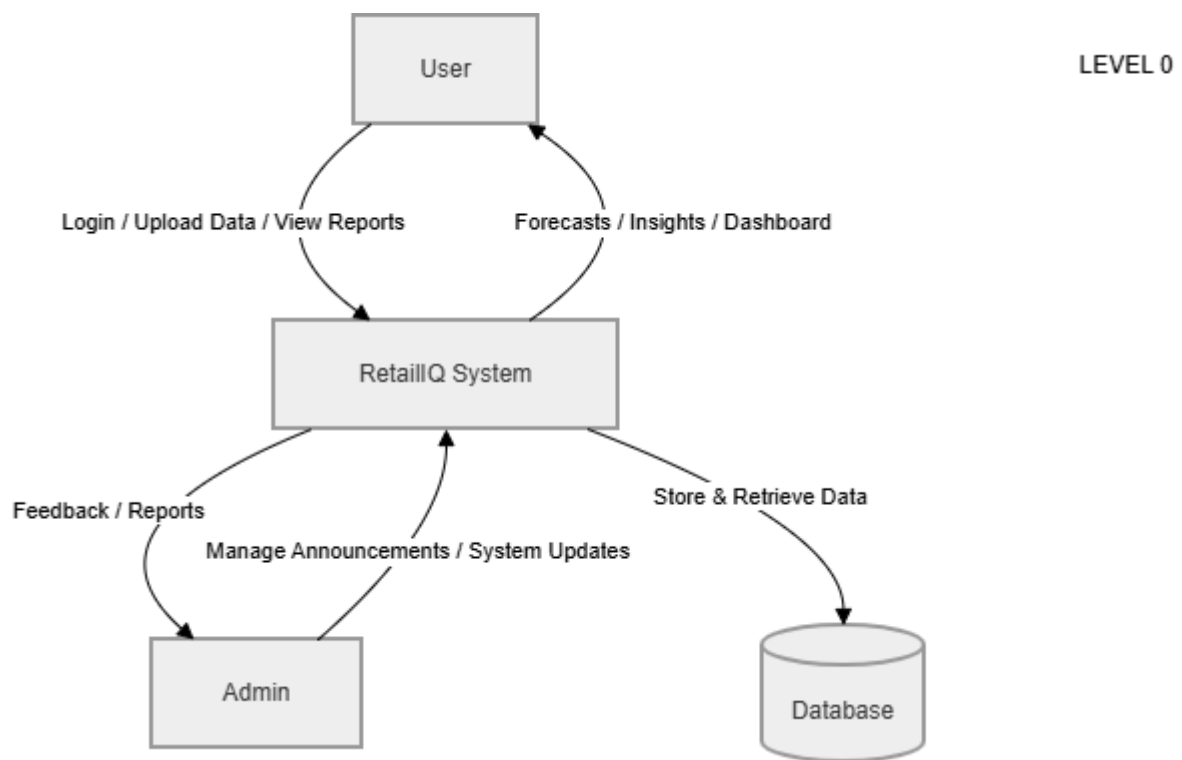
Sequence Diagram for Data Set Upload and Forecast



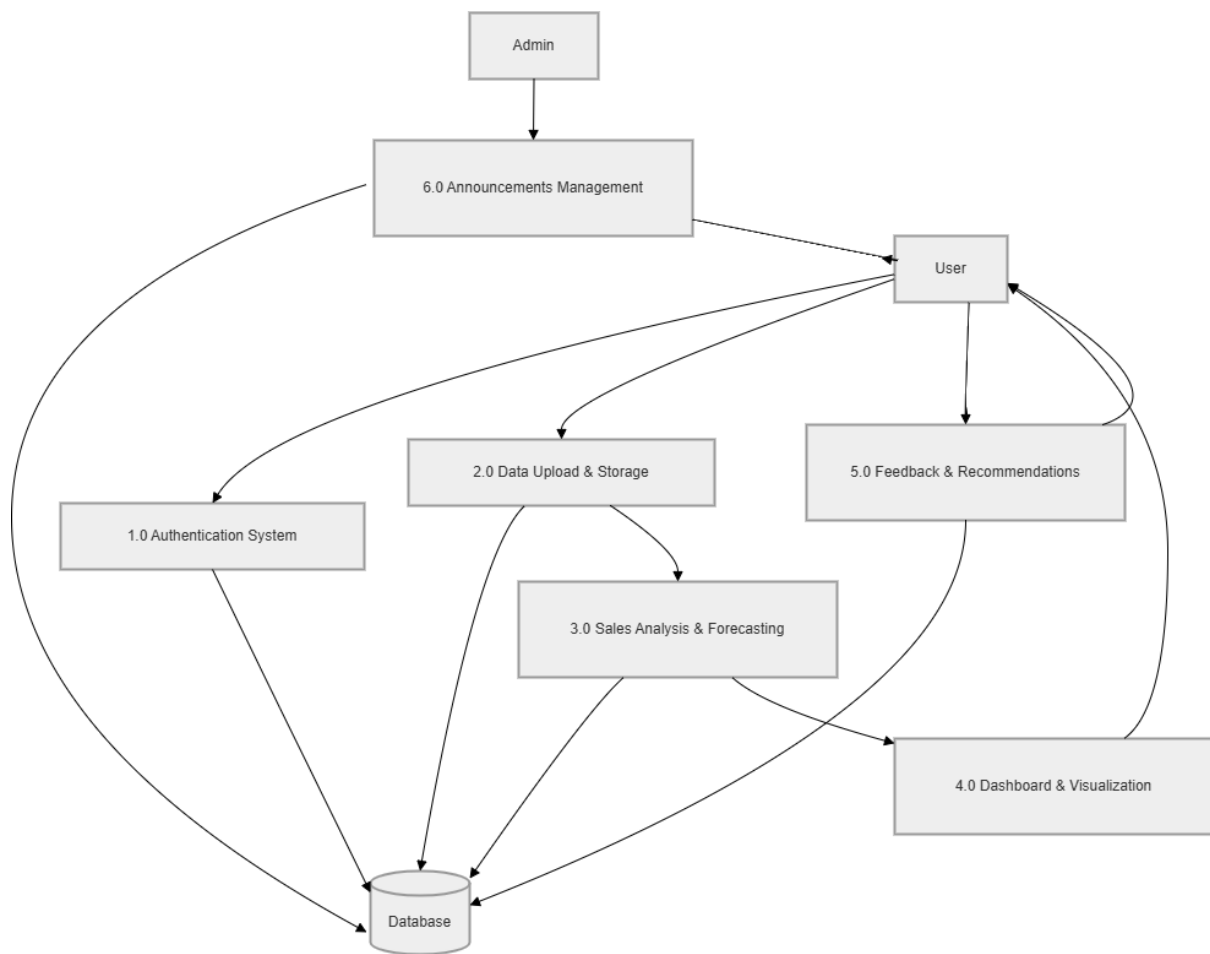
Sequence Diagram for Sales Prediction



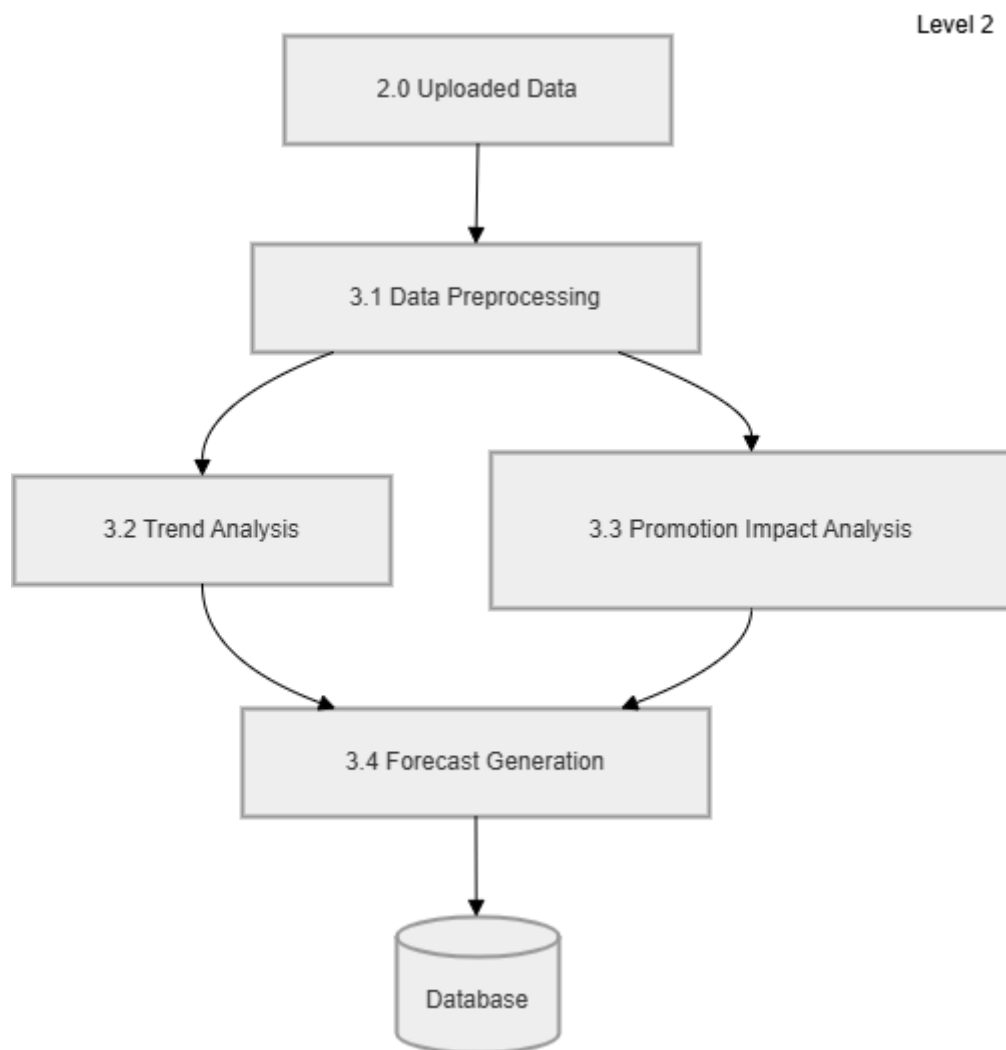
DFD Level 0



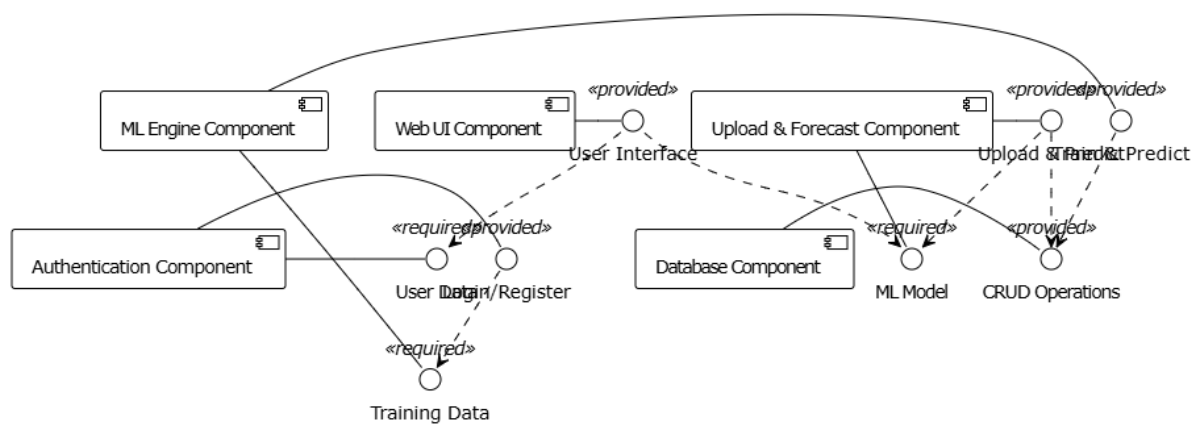
DFD Level 1



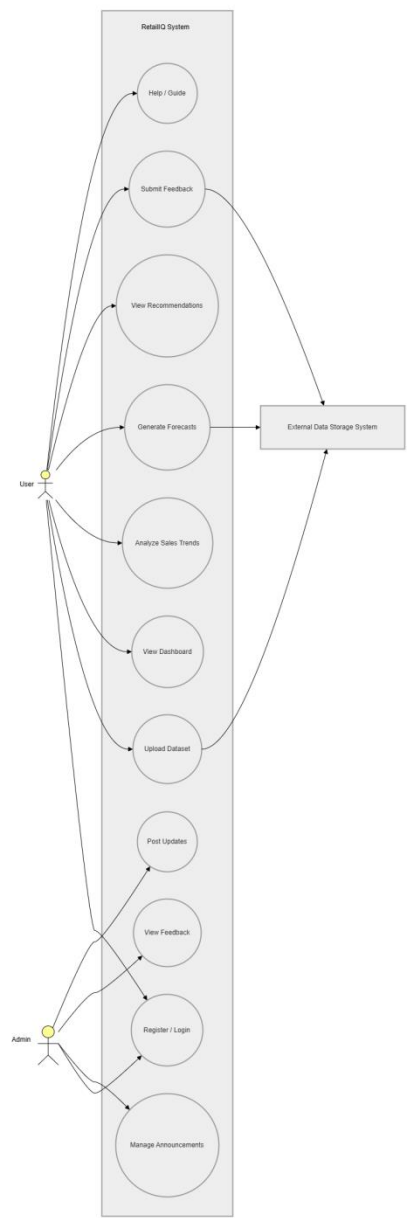
DFD Level 2



Component Diagram



Use Case



5 Work Division

The project tasks for Sprint 2 were distributed between the team members to ensure efficient completion of all activities. Both members collaborated on core machine learning tasks, while specific documentation and design responsibilities were divided individually.

Ayesha Khalid

Ayesha Khalid was primarily responsible for creating the system design and architectural documentation. Her contributions include:

- Designing **Activity Diagrams**
- Developing **System Sequence Diagrams**
- Creating **Class Diagrams**
- Designing the overall **System Architecture**
- Contributing to **model training and fine-tuning**

Uma-e-Rubab

Uma-e-Rubab focused on system analysis and supporting design documentation. Her contributions include:

- Creating **Data Flow Diagrams (DFD)**
- Developing **Use Case Diagrams**
- Designing **Component Diagrams**
- Contributing to **model training and fine-tuning**

Collaborative Work

Both team members jointly worked on:

- Training machine learning models (Linear Regression, XGBoost, LightGBM)
- Comparing model performance
- Fine-tuning selected models for improved accuracy

This division of work ensured balanced participation and effective utilization of each member's strengths.

Appendix

Work done in Sprint Along with Diagrams

```
results = {
    "Model" : ["XGBoost", "LightGBM"],
    "RMSE" : [np.sqrt(mean_squared_error(y_valid, xgb_pred)),
              np.sqrt(mean_squared_error(y_valid, lgb_pred))],
    "MAE" : [mean_absolute_error(y_valid, xgb_pred),
             mean_absolute_error(y_valid, lgb_pred)],
    "R2 Score": [r2_score(y_valid, xgb_pred),
                 r2_score(y_valid, lgb_pred)],
}

df_results = pd.DataFrame(results).sort_values("RMSE").reset_index(drop=True)
df_results.index += 1
df_results.index.name = "Rank"
print(df_results)
```

Rank	Model	RMSE	MAE	R2 Score
1	LightGBM	381.023323	134.802836	0.92095
2	XGBoost	409.056811	123.748953	0.90889

```
lgb_pred = lgb_model.predict(X_sample)
xgb_pred = xgb_model.predict(X_sample)

final_pred = 0.7 * lgb_pred + 0.3 * xgb_pred

sample_test["predicted_sales"] = final_pred
print(sample_test[["id", "predicted_sales"]])
```

	id	predicted_sales
12655	3013543	-81.344477
17039	3017927	1209.793286
2281	3003169	-4.477983
3378	3004266	1550.986731
10463	3011351	67.545389
8442	3009330	2397.596723
4405	3005293	-79.901460
17604	3018492	272.655326
14488	3015376	-15.047074
16253	3017141	731.852794